Ex. 25. ***Find the largest number of five digits which, when divided by 16, 24, 30 or 36, leaves the same remainder 10 in each case.*** **(C.P.O., 2007)**

Sol. Largest number of 5 digits = 99999. L.C.M. of 16, 24, 30 and 36 = 720.

On dividing 99999 by 720, remainder obtained is 639.

$\therefore$ Largest number of 5 digits divisible by 16, 24, 30 and 36 = (99999 – 639) = 99360.

Hence, required number = (99360 + 10) = 99370.

Ex. 26. ***Find the least number which when divided by 20, 25, 35 and 40 leaves remainders 14, 19, 29 and 34 respectively.***

Sol. Here, (20 – 14) = 6, (25 – 19) = 6, (35 – 29) = 6 and (40 – 34) = 6.

$\therefore$ Required number = (L.C.M. of 20, 25, 35, 40) – 6 = 1394.

Ex. 27. ***What is the least number which when divided by the numbers 3, 5, 6, 8, 10 and 12 leaves in each case a remainder 2 but when divided by 13 leaves no remainder?*** **(S.S.C., 2005)**

Sol. L.C.M. of 3, 5, 6, 8, 10 and 12 = 120.

So, the required number is of the form $120\,k + 2$.

Least value of k for which $(120k + 2)$ is divisible by 13 is $k = 8$.

$\therefore$ Required number = $(120 \times 8 + 2) = 962$.

Ex. 28. ***The traffic lights at three different road crossings change after every 48 sec., 72 sec. and 108 sec. respectively. If they all change simultaneously at 8 : 20 : 00 hours, then at what time will they again change simultaneously?*** **(R.R.B., 2006, M.A.T., 2005)**

Sol. Interval of change = (L.C.M. of 48, 72, 108) sec. = 432 sec.

So, the lights will again change simultaneously after every 432 seconds, i.e., 7 min. 12 sec.

Hence, next simultaneous change will take place at 8 : 27 : 12 hrs.

Ex. 29. ***Seema, Meena and Reema begin to jog around a circular stadium and they complete their revolutions in 54 seconds, 42 seconds and 63 seconds respectively. After how much time will they come together at the starting point?*** **(Bank. P.O., 2010)**

Sol. L.C.M of 54, 42 and 63 = 378.

So, the three girls will come together at the starting point in 378 seconds i.e., 6 min 18 sec.

Ex. 30. ***Arrange the fractions*** $\frac{17}{18}, \frac{31}{36}, \frac{43}{45}, \frac{59}{60}$ ***in the ascending order.***

Sol. L.C.M. of 18, 36, 45 and 60 = 180. Now,

$$\frac{17}{18} = \frac{17 \times 10}{18 \times 10} = \frac{170}{180}; \quad \frac{31}{36} = \frac{31 \times 5}{36 \times 5} = \frac{155}{180}; \quad \frac{43}{45} = \frac{43 \times 4}{45 \times 4} = \frac{172}{180}; \quad \frac{59}{60} = \frac{59 \times 3}{60 \times 3} = \frac{177}{180}.$$

Since, 155 < 170 < 172 < 177, so, $\frac{155}{180} < \frac{170}{180} < \frac{172}{180} < \frac{177}{180}$.

Hence, $\frac{31}{36} < \frac{17}{18} < \frac{43}{45} < \frac{59}{60}$.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: *Mark (✓) against the correct answer:*

1. Find the factors of 330. **(CLAT, 2010)**

(*a*) 2 × 4 × 5 × 11 (*b*) 2 × 3 × 7 × 13
(*c*) 2 × 3 × 5 × 13 (*d*) 2 × 3 × 5 × 11

2. Find the factors of 1122. **(CLAT, 2010)**

(*a*) 3 × 9 × 17 × 2 (*b*) 3 × 11 × 17 × 2
(*c*) 9 × 9 × 17 × 2 (*d*) 3 × 11 × 17 × 3

3. 252 can be expressed as a product of primes as **(IGNOU, 2002)**

(*a*) 2 × 2 × 3 × 3 × 7 (*b*) 2 × 2 × 2 × 3 × 7
(*c*) 3 × 3 × 3 × 3 × 7 (*d*) 2 × 3 × 3 × 3 × 7

4. Which of the following has most number of divisors? **(M.B.A. 2002)**

(*a*) 99 (*b*) 101
(*c*) 176 (*d*) 182

5. A number n is said to be perfect if the sum of all its divisors (excluding n itself) is equal to n. An example of perfect number is

(*a*) 6 (*b*) 9
(*c*) 15 (*d*) 21

6. $\frac{1095}{1168}$ when expressed in simplest form is